

UNIVERSITE DE LOME

Domaine: Math, Physique, Chimie

Parcours: Licence

Etablissement: Faculté des Sciences

**ANG 201: ANGLAIS SCIENTIFIQUE II**

Crédits: 2

Public cible: Etudiants

Semestre: IV

Pré-requis: ANG200

Enseignant responsable de l'UE: **YOVO K. Koboè, Docteur en Littérature de l'Afrique anglophone (ESP: Enseignant d'Anglais de spécialité); Université de Lomé: Département d'Anglais. [ged.yovo@gmail.com](mailto:ged.yovo@gmail.com)**

Disponibilité: **Mercredi 8h30-10h30**

**Mardi 9h15-11h15** (échange avec les étudiants par RESCOUL)

**Ce cours est une introduction à l'Anglais scientifique**

**A la fin de ce cours, l'étudiant doit être capable de:**

- construire de bonnes phrases en anglais en utilisant les règles de grammaire étudiées.
- utiliser correctement les mots et expressions techniques étudiés dans de bonnes phrases en Français comme en Anglais.
- répondre correctement aux questions basées sur les textes scientifiques étudiés.
- lire correctement les formules mathématiques, physiques et chimiques en Anglais.

## Syllabus

- **Study of scientific text: Natural Science**
  - Vocabulary/Pronunciation
  - Grammar: Question Tag I
  - Mathematical tools: Algebraic symbols (Part1)
- **Reading Comprehension**
  - Answers to questions based on text
  - Grammar: Question Tag II
  - Mathematical tools: Algebraic symbols (Part2)
  - Translation
- **Study of scientific text : The Pythagorean Theorem**
  - Vocabulary/ pronunciation
  - Grammar: Relative Pronouns
  - Mathematical tools: Roots and Exponents (Part1)
- **Reading comprehension**
  - Answers to questions based on text
  - Grammar: Reported Speech I
  - Mathematical tools: Roots and Exponents (Part2)
  - Translation
- **Study of scientific text: Inorganic Chemistry**
  - Vocabulary/ pronunciation
  - Grammar: Reported Speech II
  - Mathematical tools: Logarithms and Matrices
- **Reading Comprehension**
  - Answers to questions based on text
  - Grammar: English Collocations with Make and Do (Part1)
  - Mathematical Tools: Elementary Arithmetic Operations (Part 1)
  - Translation
- **Study of scientific text: Chemistry Jobs**
  - Vocabulary/Pronunciation
  - Grammar: English Collocations with Make and Do (Part2)
  - Mathematical Tools: Elementary Arithmetic Operations (Part 2)
- **Reading Comprehension**
  - Answer the questions based on the text
  - Grammar: (be)used to/get used to
  - Mathematical Tools: Association Signs, Sines, and Cosines
  - Translation
- **Study of scientific text: Applied Physics**

- Vocabulary/ pronunciation
- Grammar: The use of Unless
- Mathematical Tools: Differential and Integral Calculus
- **Reading Comprehension**
  - Answer the questions based on the text
  - Grammar: The use of both, either and neither
  - Reading Molecular Formulas and Equations (Part1)  
(MOLECULAR FORMULAS)
  - Translation
- **Study of scientific text: Physics Jobs**
  - Vocabulary/pronunciation
  - Grammar: Subject verb Inversion
  - Reading Molecular Formulas and Equations (Part2)  
(CHEMICAL EQUATIONS)
- **Reading comprehension**
  - Answers to questions based on the text
  - Grammar: Phrasal verbs
  - Reading Molecular Formulas and Equations (Part3)  
(SOME EXPRESSIONS IN CHEMISTRY)
  - Translation

**Modalités d'évaluation:** Un Devoir (de recherche) après 6 séances de cours et un Examen après 12 séances de cours.

## Bibliography

Défourneaux, Marc and Défourneaux Roldan, Michelle. (1980) *Do You Speak Science?* Bordas: Paris.

Rodney, Huddleston and Geoffrey, K. Pullum. (2002) *The Cambridge Grammar of English Language*. Cambridge: Cambridge University Press.

<https://www.britannica.com/technology/>, retrieved on 26-04-2020

<https://www.britannica.com/place/>, retrieved on 26-04-2020

<https://www.btb.termiumpius.gc.ca/tpv2guides/guides/wrtps/> retrieved on 29/04/2020

<https://www.eslbase.com/grammar/> retrieved on 29/04/2020

© 2014 [www.perfect-english-grammar.com](http://www.perfect-english-grammar.com) retrieved on 28/04/2020

<https://www.bbc.co.uk/learningenglish/course/towards-advanced/> retrieved on 30/04/2020

## Session1-Study of scientific text: Natural Science

### VOCABULARY:

**Explain the underlined words through definition, synonym, antonym or translation.**

eg.

Milky Way galaxy: a large spiral system consisting of several hundred billion stars, one of which is the sun

Haber-Bosch process: also called Haber ammonia process, or synthetic ammonia process is a method of directly synthesizing ammonia from hydrogen and nitrogen from the air (developed by the German physical chemist Fritz Haber)

To hamper: prevent something from happening or progressing normally

To underlie: to serve as a basis of, to form the foundation of

....etc

### **Text:**

Natural science is one of three divisions of science, the other two being the social sciences-psychology, sociology, economics, etc.-and the formal sciences-mathematics, logic, and statistics. The natural sciences are astronomy, biology, chemistry, earth science, and physics. There are also **cross-disciplines** such as biophysics which **integrate** between sciences. Natural science has been practiced for millennia, but prior to the 17th century it was called natural philosophy and was less scientific.

The introduction of the **scientific method** by Sir Francis Bacon and the **ensuing** scientific revolution came to create what is considered science to day. The natural sciences, especially physics, are often referred to as **«hard science»** because of the **heavy** use of objective data and quantitative methods. In contrast, the social sciences rely more on qualitative evaluations and thus tend to have less certain conclusions. The study of human beings in the social science is **hampered** both by the complexity of humans and the increased tendency towards unscientific bias when studying humans as opposed to, say, rocks. Much of what defines our modern civilization are advances in knowledge and technology caused by investigations in the natural sciences. For instance, the production of most if not all of the food you eat can be traced back to a chemical process called the **Haber-Bosch process** which was developed during WWI. This chemical process allows the creation of fertilizer nitrates from atmospheric nitrogen, rather than relying on biologically fixed sources of nitrogen such as cow dung or bat guano. Thanks to developments in biology, especially accelerating in the last half-century, we have advanced medicines capable of curing or treating many diseases that were otherwise fatal. The **scourges** of the 19th century and before, such as **plague** and **smallpox**, are now largely

under control or eliminated thanks to biology research. The biological subfield of genetics has allowed us to understand the very code of life itself, and recognize the way it expresses itself within each of us. Advances in earth science have allowed us to extract huge amounts of minerals and petroleum from the Earth's crust, powering the engines of modern civilization and industry to the present day. Paleontology, a subfield of Earth science, has given us a window into the Earth's distant past, long before humans ever existed. This helps us trace the reasons why the spectrum of life and the geology of Earth is the way it is today. Some of the most fascinating of the natural sciences are physics and astronomy, the most quantitative fields in all of empirical science. Physics is the science that **underlies** all others, and has been subject some of the most unexpected twists and turns in the last century, most notably the discovery that matter and energy, space and time are each facets of each other. Astronomy has led us to discover an enormous amount about the universe we live in. Up until about a century ago, it was thought that the entire universe was just the Milky Way galaxy, until a series of debates and observations clued us in that it was literally millions of times bigger than we previously imagined. *The natural sciences are one of the most important and interesting fields of human endeavor, and while the specifics of future advances are largely unpredictable, one thing is for sure: scientific findings will continue to improve our lives and fire up our imaginations for a long time to come.*

### Grammar: Question Tags (Part 1)

Question tags are made of an **auxiliary verb** plus a **pronoun**, which go at the end of a sentence, either to ask for more information about something, or to confirm something which we believe is true. The auxiliary verb matches the tense of the main verb and the pronoun comes directly from the noun.

Question tags usually work in opposition. If the main sentence is affirmative, then the question tag is usually negative, so: **You do live here, don't you?** and vice versa.

Eg. – Nadia sells oranges in the market, **doesn't she ?**

- Koboè went to Munich last week, **didn't he ?**

- The people do not believe in God, **do they ?**

- Sandra hasn't seen the teacher coming, **has she ?**

#### 1. Negative or Limiting Adverbs

Sentences which use a negative or limiting adverbs (**never, no, hardly, scarcely, little**, etc.) are treated as negative by the question tag, even though their construction is positive.

Therefore the question tag is positive.

eg. **They never go on holiday, do they?**

## 2. Indefinite pronouns

Sentences with indefinite pronouns such as **someone**, **anyone**, **no one** and **everyone**, use **they** in the tag.

Eg. - **No one** cares, **do they**?

- **Everyone** left, **didn't they**?

- **Someone's** at the door, **are they**?

With other indefinite pronouns such as **something** and **everything**, use **it**:

eg. - **Everything** is OK, **isn't it**?

- **Nothing** matters, **does it**?

- **Something** smells bad, **doesn't it**?

## 3. Imperatives

Imperatives are commands, suggestions, offers, advice etc. They are constructed of the bare infinitive in the affirmative and use **don't** in the negative: **Sit down! Don't sit there!**

Imperatives don't have a tense or use an auxiliary verb in the same way as other sentences do. To make a question tag with imperatives, we use **won't you?** although other modal verbs can be used, such as **will**, **would**, **can**, and **could**.

Eg. - **Sit down**, **won't you**?

- **Open the window**, **would you**?

- **Don't go outside**, **will you**?

- **Keep quiet**, **won't you**?

## Mathematical tools: ALGEBRAIC SYMBOLS (Part1)

1- = equals or is equal to sign

2-  $\approx$  *approximately equal to sign*

3-  $>$  greater than sign

4-  $<$  *less than sign*

5-  $\geq$  *greater than or equal to sign*

6-  $\leq$  *less than or equal to sign*

7-  $a < x \leq b$  *x is greater than a and less than or equal to b*

NB: Words like Superior or Inferior are unacceptable for 3 to 7.

8-  $x^*$  x star

9-  $x'$  x prime

10-  $x''$  x double prime

## Session2-Reading Comprehension

**Read the text carefully and answer the questions.**

### QUESTIONS

- 1°) How was natural science called before the 17th century?
- 2°) What brings about the natural science?
- 3°) Say whether the following statements are true or false: a-Physics is a natural science.  
b-Biology is a natural science d-Chemistry is a natural science d-Geology is a natural science
- 4) How will you differentiate between Natural sciences and Social sciences?
- 5)-State the importance of Natural sciences in human beings' life

### Grammar: Question Tags (Part 2)

#### 4. Suggestions with Let's:

When making a suggestion, it is common to use **let's**. **Let's** stands for **let us**, for example: **Let's go to the cinema**. When we use let's in a question tag we always use **shall we**, regardless of whether **let's** is affirmative or negative. So for example:

**Let's go to the cinema, shall we?**

**Let's not go to the cinema, shall we?**

#### 5. Double positives

Double positives are possible. This is where the both the sentence and the question tag are positive, for example:

**You're going to become a pilot, are you?**

Double positives are common when people are reacting to news, repeating something they have just heard or reacting in an emotional way to something. For example:

**Eg. - You're getting married, are you?**

**- You've lost your wallet, have you?**

#### 6. Opinions

If you start a sentence with **I think**, don't use the question tag **do/don't I**. Make the question tag agree with the main information in the sentence. This does not apply in the second or third person.

**Eg. - I think it's a nice day, isn't it?**

**- I don't think that's a good idea, is it?**

**- You thought you'd be OK, did you?** (second person)

**- He thinks he's going to university, does he?** (third person)

However, in some situations, for example, when we're being sarcastic, we do make the tag agree with I think.

A: **Yes, yes. Well done. I know you think you are so clever.**

B: **Oh! So, I think I'm clever, do I? Not at all.**

### Mathematical tools: ALGEBRAIC SYMBOLS (Part2)

11-  $x_c$   $x$  sub  $c$

12-  $x^b$   $x$  super  $b$

13-  $|a - b|$  distance from  $a$  to  $b$

14-  $|x|$  the absolute value of  $x$

15-  $-x$  negative of  $x$

16-  $\frac{1}{x}$  the reciprocal of  $x$

17-  $\sum_{i=1}^n x_i$  Summation from 1 to  $n$  of  $x$  sub  $i$

18-  $\prod_{i=1}^n x_i$  Product from 1 to  $n$  of  $x$  sub  $i$

TRANSLATION: Translate the passage in italic into French

## Session3-Study of scientific text: The Pythagorean Theorem

### VOCABULARY:

**Explain the underlined words through definition, synonym, antonym or translation.**

eg.

to name after sb/sth: to give someone or something the same name as another person or thing (eg. This man named all his children after footballplayers.)

to come up with sth = to discover/ find/ invent sth

to predate: to have existed or happened before another thing ( $\neq$  antedate)

to cut accross: to go across something in order to make your route shorter

to skirt: to go around the edge of a place or thing (eg. They skirted the field to avoid the mud)

square root: racine carrée

to give (some) one credit: To give someone praise or recognition.

...etc

### **Text:**

The Pythagorean theorem is a mathematical theorem **named after** Pythagoras, a Greek mathematician who lived around the fifth century BCE. Pythagoras is usually **given the**

**credit** for coming up with the theorem and providing early proofs, although evidence suggests that the theorem actually predates the existence of Pythagoras, and that he may simply have popularized it. Whoever deserves the credit for developing the Pythagorean theorem would undoubtedly be pleased to know that it is taught in geometry classes all over the world, and it is **utilized** on a daily basis for everything from doing high school math homework to making complex engineering calculations for the Space Shuttle. ***According to the Pythagorean theorem, if the lengths of the sides of a right triangle are squared, the sum of the squares will equal the length of the hypotenuse squared.*** This theorem is often expressed as a simple

formula:  $a^2 + b^2 = c^2$ , with  $a$  and  $b$  representing the sides of the triangle, while  $c$  represents the hypotenuse. In a simple example of how the Pythagorean theorem might be used, someone might be **wondering** about how long it would take to cut across a rectangular lot of land, rather than skirting the **edges**, relying on the principle that a rectangle can be divided into two simple right triangles. He or she could measure two adjoining sides, determine their squares, add the squares together, and find the square root of the sum to determine the length of the lot's diagonal. Like other mathematical theorems, the Pythagorean theorem relies on proofs. Each proof is designed to create more supporting evidence to show that the theorem is correct, by demonstrating various applications, showing the shapes that the Pythagorean theorem cannot be applied to, and attempting to disprove the Pythagorean theorem to show, in reverse, that the logic behind the theorem is **sound**. Because the Pythagorean theorem is one of the oldest math theorems in use today, it is also one of the most heavily proved, with hundreds of proofs by mathematicians throughout history adding to the body of evidence which shows that the theorem is valid. Some special shapes can be described with the Pythagorean theorem. A Pythagorean triple is a right triangle in which the lengths of the sides and hypotenuse are all whole numbers. The smallest Pythagorean triple is a triangle in which  $a=3$ ,  $b=4$ , and  $c=5$ . Using the Pythagorean theorem, People can see that  $9 + 16 = 25$ . The squares in the theorem can also be literal; if one were to use every length of a right triangle as the side of a square, the squares of the sides would have the same area as the square created by the length of the hypotenuse. One can use this theorem to find the length of any unknown segment in a right triangle, making the formula useful for people who want to find the distance between two points. If, for example, one knows that one side of a right triangle is equal to three, and the hypotenuse is equal to five, one knows that the other side is four long, relying on the well-known Pythagorean triple discussed above.

## Grammar: Relative Pronouns, who, whom, whose, which, that

A relative **pronoun** is one which is used to refer to **nouns** mentioned previously, whether they are people, places, things, animals, or ideas. Relative pronouns can be used to join two sentences. The three most common relative pronouns are **who**, **which** and **that**. **Who** has two other forms, the object form **whom** and the possessive form **whose**.

- **Who** and **whom** are used mainly for people.

Eg.

-The musician **who** wrote this song is Canadian.

-The witnesses **whom** I interviewed gave conflicting evidence.

- **Whose** can be used for people, animals or things:

Eg.

-The man **whose** daughter won the tournament is a tennis coach.

-A dog **whose** owner lets it run loose may cause an accident.

-The tree **whose** branches shade my kitchen window is an oak.

- **Which** is used for animals in general or things.

Eg.

-Bridget visited the park with her dog, **which** likes to chase squirrels.

-Jason bought the top hybrid car, **which** will help him save on gas.

- **That** can be used for people, animals or things.

Eg.

-The musician **that** won the award is Canadian.

-The car **that** Jason bought runs on electricity and gas.

-The dog **that** is chasing the squirrels belongs to Bridget.

## SUMMARY

|              | Subject | Object | Possessive     |
|--------------|---------|--------|----------------|
| Personal     | Who     | Whom   | Whose          |
| Non-Personal | Which   | Which  | Whose/of which |

## Mathematical tools: ROOTS AND EXPONENTS (Part 1)

(a) **The Root of a Number (la racine d'un nombre)**

1-  $\sqrt[n]{x}$  the nth root of x

2-  $n=2$   $\sqrt{x}$  the square root of x or the root of x

3-  $n=3$   $\sqrt[3]{x}$  the cube root of x

From  $n=4$  onward we say the fourth root, the fifth root, the sixth root, the hundredth root etc

- 4-  $n=4$   $\sqrt[4]{x}$  the fourth root of x
- 5-  $n=5$   $\sqrt[5]{x}$  the fifth root of x
- 6-  $n=10$   $\sqrt[10]{x}$  the tenth root of x
- 7-  $n=100$   $\sqrt[100]{x}$  the hundredth root of x

### Example Problems

- 1-  $y = \sqrt{x}$  y equals or y is equal to the (square) root of x
- 2-  $y = \sqrt[m]{x}$  y equals the mth root of x
- 3-  $y = \sqrt[3]{x}$  y equals the cube root of x
- 4-  $y = \sqrt[10]{x}$  y equals the tenth root of x

## Session4-Reading comprehension

**Read the text carefully and answer the questions.**

### QUESTIONS:

- 1) a-Who is the author of the Pythagorean Theorem? b-Give his nationality and the time of his theorem.
- 2) This theorem is used only in Anglophone countries. True/False
- 3) State the theorem in English and French.
- 4) What can one do with that theorem?

### Grammar: THE REPORTED SPEECH (Part 1)

It is a grammatical structure which consists in telling or accounting for something another person said directly some time ago, recently or at a time far back in the past. In the reported speech (indirect speech) the speaker tells of an event or a statement someone talked about or put before. Because the statement he is making is not a repetition but a new statement, there is a series of changes based on the nature of the reporting verbs, the sentences, the person, time, and place that we need to consider when turning sentences from the direct to the indirect speech.

#### The Nature of the Sentences

Turning sentences from the direct speech to the indirect speech requires a clear identification of the type of sentences we have to work on. Students or language users must know if the

sentence is a statement (a declarative sentence), a question (interrogative sentence) or an order (imperative sentence). This first step orients the various changes in reported sentences.

### Presentation

He said, "I like this song." → He said that he liked that song.

Daddy said : "I came yesterday" → Daddy said that he had come the previous day.

Abalou said : "Next week I will go to Kara" → Abalou said that he would go to Kara the following week.

### THE OCCURRENCE OF CHANGES

The changes often concern pronouns (personal, possessives, reflexives, and demonstratives), adverbs (of place and time) and tenses.

#### 1- CHANGE OF PRONOUNS

##### a- Personal Pronouns

| Direct Speech         |         | Indirect Speech        |
|-----------------------|---------|------------------------|
| I, you, he, she, it   | Becomes | He, she, it (I, you)   |
| We, you, they         |         | They (we, you)         |
| Me, you, him, her, it |         | Him, her, it (me, you) |
| Us, you, them         |         | Them (us, you)         |

##### b- Possessives

###### ✓ Adjectives

| Direct Speech           |         | Indirect Speech           |
|-------------------------|---------|---------------------------|
| My, your, his, her, its | Becomes | His, hers, its (my, your) |
| Our, your, their        |         | Their (our, your)         |

###### ✓ Pronouns

| Direct Speech          |         | Indirect Speech        |
|------------------------|---------|------------------------|
| Mine, yours, his, hers | Becomes | His, her (mine, yours) |
| Ours, yours, theirs    |         | Theirs (ours, yours)   |

###### ✓ Reflexives

| Direct Speech                              |         | Indirect Speech                               |
|--------------------------------------------|---------|-----------------------------------------------|
| Myself, yourself, himself, herself, itself | Becomes | Himself, herself, itself (myself, yourselves) |
| Ourselves, yourselves, themselves          |         | Themselves (ourselves, yourselves)            |

###### ✓ Demonstratives

| Direct Speech |  | Indirect Speech |
|---------------|--|-----------------|
|---------------|--|-----------------|

|       |         |       |
|-------|---------|-------|
| This  | Becomes | That  |
| These |         | Those |

## 2- CHANGE OF ADVERBS

The change of adverbs concerns adverbs of time and place. And these changes describe the possible movements in space or the passing of time before the reported speech is built by the one reporting. The following changes occur with adverbs of time and place:

|          |   |                    |
|----------|---|--------------------|
| Here     |   | there              |
| Now      |   | then               |
| Ago      |   | before             |
| Today    | → | that day           |
| Tonight  |   | that night         |
| Tomorrow |   | the following/next |

|                        |                |                    |
|------------------------|----------------|--------------------|
| last (week, day, year) |                | the week before    |
| yesterday →            | the day before |                    |
| next (week, day, year) |                | the following week |

✓ Other changes

|         |   |                |
|---------|---|----------------|
| Ago     | → | before         |
| Thus    | → | so             |
| At once | → | at that moment |
| Now     | → | then           |

## 3- CHANGE OF TENSES

The change of tenses relates to the time that has just gone between the moment the direct speech has been produced and the moment the one reporting has talked.

|         | Direct Speech         | Indirect Speech           |
|---------|-----------------------|---------------------------|
| Present | Simple present        | Simple past               |
|         | Present perfect       | Past perfect              |
|         | Present progressive   | Past progressive          |
|         | Present perfect prog. | Past perfect prog.        |
| Past    | Simple past           | Past perfect              |
|         | Past perfect          | Past perfect              |
|         | Past progressive      | Past perfect prog.        |
|         | Past perfect prog.    | Past perfect prog.        |
| Future  | Simple future         | Conditional               |
|         | Future perfect        | Conditional perfect       |
|         | Future perfect prog.  | Conditional perfect prog. |
| -----   | Conditional           | Conditional               |
| -----   | Imperative            | Infinitive                |

### Imperative infinitive

"Open the door at once" → He told me to open the door at that moment.

Negative imperative negative infinitive

"Don't greet her anymore" → He told him not to greet her anymore.

### Mathematical tool: ROOTS AND EXPONENTS (Part 2)

#### (b) *Exponents; Raising a Number to a Power (elever à la puissance)*

A number is raised to the nth power or to the power n

1)-  $x^n =$  x to the power n or x to the nth power or x raised to the nth power

2)- For  $n=2$   $x^2 =$  x squared

3)- For  $n=3$   $x^3 =$  x cubed

4)-  $n=4$   $x^4 =$  x to the power four or x to the fourth power or x raised to the fourth power

5  $n=5$   $x^5 =$  x to the power five or x to the fifth power or x raised to the fifth power

6  $n=9$   $x^9 =$  x to the power nine

7  $n=26$   $x^{26} =$  x to the power twenty six

#### Example problems

1  $a^x =$  a to the power x or a to the xth power

2  $a^7 =$  a to the power seven or a to the seventh power

3  $a^{-x} =$  a to the power negative x

4  $\frac{1}{2}bm^2 =$  one half b m squared

5  $\frac{4}{3}m^2y^4 =$  four thirds m squared y to the power four

6  $(a + b + c)^y =$  a plus b plus c **all** to the power y

7  $\frac{1}{2}y^2 =$  one half y squared

#### Case of Fractional Exponents

1  $a^{\frac{1}{2}} =$  a to the power one-half

2  $x^{\frac{3}{4}} =$  x to the power three quarters

3  $x^{\frac{-3}{4}} =$  x to the power minus (negative) three quarters

4  $a^{\frac{m}{n}} =$  a to the power mnths or a to the power m over n

TRANSLATION: Translate the passage in italic into French

## Session5-Study of scientific text: Inorganic Chemistry

### VOCABULARY:

**Explain the underlined words through definition, synonym, antonym or translation.**

eg.

Misconception: a wrong or inaccurate idea or conception

To veer away (from): stay away from, avoid/change heading or direction

assumption: something that you accept as true without question or proof

to blur: to make or become unclear or less distinct. (eg. Tears blurred her vision)

...etc

**Text:**

*Inorganic chemistry is a branch of chemistry that deals with the properties and behavior of inorganic compounds. Inorganic compounds are generally those that are not biological, and characterized by not containing any hydrogen and carbon bonds.* It is almost easier to discuss inorganic chemistry in terms of what it is not-organic chemistry. Organic chemistry is the study of any chemical reaction that involves carbon, which is the element that all life is based on. The term organic has traditionally referred only to animal or plant matter. Therefore, there is a common **misconception** that organic chemistry always refers to life processes, or that inorganic chemistry applies to everything that does not. This **assumption** is **inaccurate**. Many chemical processes **veer away from** this line of thinking. There are many chemical life processes that depend on inorganic chemical processes. There are exceptions to every rule. Although carbon is the main common element in organic chemistry, inorganic chemical compounds can contain carbon, too. For example, carbon monoxide and carbon dioxide both contain carbon, but are inorganic compounds. Carbon dioxide, in particular, is also very important to chemical processes necessary for life, especially plant life. The truth is that the lines between inorganic and organic chemistry are often **blurred**. There are many branches of inorganic chemistry available for specialization. Geochemistry is the study of the chemicals of the Earth and other planets. It covers the chemical compositions of rocks and soil. Within the field of geochemistry, there are several subfields. These subfields include isotope geochemistry, cosmochemistry and biogeochemistry. Another type of inorganic chemistry is physical chemistry, which relates to the concept of physics in chemical systems. This field is also sometimes called physicochemistry. It uses the principles of

thermodynamics, quantum chemistry and kinetics as its basis. On the other hand, bioinorganic chemistry is the study of compounds containing metal-carbon bonds within biological systems. This is a particularly interesting branch of inorganic chemistry because it also **incorporates** aspects of organic chemistry into it. Bioinorganic chemistry focuses on the presence of metal ions in biochemical processes. Inorganic chemistry **lends** itself to many different industries, including education, environmental science, and government agencies. A scientist who focuses on inorganic chemistry might create or improve formulas for household cleansers. He may also work in chemical research, **coming up** with new ways to manipulate the properties of metallic elements into useful functions.

## Grammar: THE REPORTED SPEECH (Part 2)

### REPORTED QUESTIONS

The reporting of questions is done according to the type of questions asked namely the auxiliary question or Yes/No Questions and the "WH/how Questions.

#### 1-Auxiliary/ Yes or No Questions

The reported speech of these questions requires the use of "IF" or "WHETHER".

e.g. - "Will you come to the club meeting?", She asked him → She wanted to know if he would come to the meeting.

- "May I talk to you, please?" → He asked me if he might talk to me.

#### Practice with the following questions:

"Shall we be waiting for the contractors?", they asked me

"Could one see the waves from here?", he wanted to know

"Is Didier your brother?", she asked Koni

#### 2- WH" and "How" Questions:

In the reported speech of these questions, the question words are used before the subject of the sentence.

e.g. "When is the party?", she asked me → She asked me when the party was.

"Who is this man?", they asked → They wondered/wanted to know who that man was.

#### Practice with the following questions:

"How did you achieve the qualities you have?"

"What is the time?"

#### Note:

##### 1- There is no change in tense when:

**A-The sentence in the direct speech states a general truth.**

eg.

They said "The sun rises in the east and sets in the west." → They said that the sun rises in the east and sets in the west.

The biologist said "Only women bear pregnancies." → The biologist said that only women bear pregnancies.

### **B-The reporting verb is not conjugated in the past.**

eg.

She says "I am happy" → She says that she is happy.

I have declared "We shall overcome this system." → He has declared that they shall overcome that system.

2- Sentences in the reported speech do not bear the quotation marks.

3- The reported question does not take an interrogation mark at the end of the sentence.

### **SPECIAL CASES**

Some sentences embody two statements, or statements and questions. The reporting of such sentences requires an identification of the reporting verb and then its use before the question or statement.

eg.

"I will call her first and she will send a copy of the contract." → He said he would call her first and she would send a copy of the contract.

"My name is Karou. What is your name." → He said his name was Karou and asked me what my name was.

- "Are you done? I am about to finish." → He asked me if I was done and told me he was about to finish.

## **Mathematical tool: LOGARITHMS AND MATRICES**

### **Logarithms**

- 1-  $\log_a x$  = log x to base a
- 2-  $\ln x$  = natural log of x
- 3- e base of natural logarithm
- 4-  $e^x$  e to the x
- 5-  $e^{f(x)}$  = exponential f of x

### **Matrices**

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix}$$

We say that A is a matrix with 2 rows and 3 columns

or we say A is a 2 by 3 matrix

## Session6-Reading Comprehension

Read the text carefully and answer the questions.

### QUESTIONS:

- 1) What is inorganic chemistry?
- 2) What is the misconception about inorganic chemistry?
- 3) All the chemical compounds that contain carbon are inorganic. True/False?

### Grammar: ENGLISH COLLOCATIONS WITH make and do (Part1)

When two or more words are used together and they sound natural they are said to collocate; (to collocate = vivre ensemble).

DO and MAKE are verbs and in the example sentences that will be given, they can be used in all tenses.

### English Collocations with 'DO'

**'DO'** is used to describe actions, obligations and repetitive tasks (tasks that we do often)

Do the housework = faire le ménage                      Do the dishes = laver les vaisselles

Do homework =faire le devoir                              Do business = faire les affaires

Do a course = suivre un cours                              Do exercise = faire l'exercice

Do your hair = se coiffer/faire les cheveux              Do one's nails = faire ses ongles

Do your/my best = faire de votre mieux

Do something/nothing/everything/anything = faire quelquechose/ne rien fait/il faux tout faire/faire n'importe quoi

Do badly = faire mal/avoir de mauvais résultats/se trouver en difficulté

Do good = faire du bien

### EXAMPLES

1- **do the housework Yesterday** eg.Ali was too tired to do the housework.

2- **do the dishes** eg.Afi made dinner and Koffi did the dishes afterwards.

3- **do homework** eg.There is no TV until you have done your homework.

4- **do business** eg.My father does business in Lome and Tsevie

5- **do a course** eg.My son is doing a physics course at University.

6- **do exercise** eg.I do one hour of exercise every day.

7- **do your hair** eg.My sister spends too much time doing her hair.

8- **do your nails** eg.She did her nails five minutes ago. They are still wet.

9- **do something/nothing/everything/anything.**

Eg.I have done something bad to my brother. I must apologize.

10- **do badly** eg.We all did badly in the exam. The highest score was 11.

11- **do good** eg.It pays to do good in life.

## Mathematical tools: ELEMENTARY ARITHMETIC OPERATIONS (Part 1)

### Symbols

1- = is equal to or equals sign

2- - the minus sign

3- + the plus sign

4-  $\times$  the multiplication sign

5-  $\div$  the division sign

6-  $\pm$  the plus/minus sign

### (a) Addition

$a+b=c$  a plus b equals c (a plus b is equal to c)

If you add a to b you obtain a sum c

### Expressions with add and sum

to add up to = s'elever à

to sum up = résumer

### Example problem

$32+12=44$  thirty-two plus twelve equals forty-four

**TRANSLATION:** Translate the passage in italic into French

## Session7-Study of scientific text: Chemistry Jobs

### VOCABULARY:

Explain the underlined words through definition, synonym, antonym or translation.

eg.

Manufacturing plant: a factory or building where goods are manufactured

To hire: employ (someone) for wages

Forensic Chemistry: Apply scientific disciplines to physical evidence (Forensic chemists apply knowledge of chemistry to legal issues, usually by analyzing physical evidence to assist police investigators in solving crimes. Most forensic chemists work for law enforcement agencies at the local, state and federal levels.)

...etc

**Text:**

## CHEMISTRY JOBS

There are several ways to look at the different types of chemistry jobs. One can look at the main branches of chemistry, as well as the different industries that chemists work in. One can also look at the specialized careers in which chemistry jobs are offered. Chemists may choose to study analytical chemistry, biochemistry, inorganic chemistry, organic chemistry, to name some of the main areas of the field. Analytic chemists reveal the composition and structure of substances. Biochemists study living organisms and their chemical composition.

*Inorganic chemists focus on compounds composed of elements other than carbon, while organic chemists analyze the chemistry of carbon compounds, that is, the compounds that make up living things. Chemistry jobs are offered in some standard locations. Chemistry work is often done in laboratories, as one might expect.* But chemists may also work in offices, schools, manufacturing plants, and sometimes **out doors** if, for example, their specialty **involves** the environment or water. One area for chemistry jobs is in academics. High schools, community colleges, four-year colleges, universities, and online programs **employ** chemists to teach chemistry and its subspecialties. Instructors in a four-year college or university program will probably need a Ph.D., but the requirements may be less for online instruction, community colleges, and high schools. A high school teacher may be **hired** with only a bachelor of Science degree in chemistry. Chemists may also work in research and development, investigating the properties of matter and its behaviour and working to create new products and processes, as well as synthesize products that **occur** naturally. Some chemistry jobs are in manufacturing plants in which production and quality control are both areas in which chemistry is combinatorial. Chemistry jobs, for example, are involved with testing and analysis, while medical chemistry jobs are involved with identifying the structure of compounds in the service of the pharmaceutical industry. Chemistry jobs that involve other skill sets in addition to chemistry are also available including jobs in science writing, sales, and consulting. According to the American Chemical Society, industry specialties for Chemists include food and flavour chemistry, chemical technology, chemical engineering, oil and petroleum chemistry, polymer chemistry, pulp and Paper chemistry, textile chemistry, and water chemistry. Chemists also go on to take positions as forensicchemists, environmental

chemists, and geochemists. Chemists may also add credentials to be able to work as Pharmacists, toxicologists, medical doctors, and biochemists.

### Grammar: English Collocations with 'MAKE'

'**Make**' is used to describe actions in many areas of life: food, money, relationships, communications, plans and progress...

**Make breakfast** = faire le petit déjeuner

Make a cup of tea = faire une tasse du thé

Make money = faire de l'argent

Make a profit = réaliser de bénéfice

Make fun of someone = se moquer de quelqu'un(e)

Make friends = se faire des amis

Make do (make somebody do something = obliger quelqu'un de faire quelque chose

Make a phone call = faire un coup de fil

Make a promise = faire une promesse

Make a suggestion/an observation/an excuse/a complaint = faire une suggestion/une observation/s'excuser/porter plainte/

Make a mistake = commettre une erreur

Make plans = se proposer de faire

Make up your mind = prendre la décision, décider de faire qch

Make progress = faire de progress

### EXAMPLES

#### **make breakfast**

Afi makes breakfast for the family every morning.

- 1 **make a cup of tea/coffee** eg. Would you like to make a cup of tea or coffee?
- 2 **make money** eg. I don't make much money in my job
- 3 **make a profit** eg. My company made a profit in the second year of operation.
- 5 **make fun of someone** eg. Ali makes fun of his sister at home.
- 6 **make friends** eg. At school Koffi makes friends easily.

7 **make up** = résoudre un problème entre amis/frères/parents.....

eg. Ali and Koffi made up after insulting each other.

8 **make a phone call** eg. Please wait a minute ; I need to make a phone call

9 **make a promise** eg. My brother made many promises but broke them all.

10 **make a suggestion/an observation/an excuse/a complaint**

eg. Afi made many good suggestions at the meeting.

11 **make a mistake (in your calculation)**

eg. I made many mistakes in the math exam.

12 **make plans** eg. I am making plans for my sister's marriage next week.

13 **make up your mind**

eg. Ali has made up his mind to study Mathematics at University.

14 **make progress (in your studies)**

eg. We have made little progress. This problem is difficult to solve.

## EXERCISES

Fill the gaps with MAKE OR DO in the correct form (tense).

- 1 The company ..... a lot of business with Japan.
- 2 I think you've ..... a mistake.
- 3 They're ..... plans for the wedding next month.
- 4 It is easier to ..... money in some businesses than in others.
- 5 The doctor told me to ..... exercise regularly.
- 6 John ..... me a cup of coffee yesterday.
- 7 Who is going to ..... breakfast? You or me.
- 8 He ..... a lot of money last year.
- 9 You must always ..... your best.
- 10 Next week I ..... all the students ..... their homework.

## Mathematical Tools: ELEMENTARY ARITHMETIC OPERATIONS (Part 2)

### (b) Subtraction

$a-b=c$  a minus b equals c

If you subtract b from a you obtain the difference c

Example problem

$53-30=23$  fifty-three minus thirty equals twenty-three

**(c) Multiplication**

$a \times b = c$  a multiplied by b equals c; alternatively we say a times b equals c  
If you multiply a by b you get c

Example problems

$15 \times 12 = 180$  fifteen multiplied by twelve equals one-hundred and eighty

**(d) Division**

$a \div b = c$  a divided by b equals c; alternatively we say a over b equals c;  
If you divide a by b you get c

Example problem

$5 \div 2 = 2.5$  five divided by two equals two point five

**Session 8-Reading Comprehension**

**Read the text carefully and answer the questions.**

**QUESTIONS**

- 1). State the jobs a chemist can do.
- 2). Mention some of the sub-branches of chemistry that appear in the text.
- 3). What qualifies one to teach chemistry in high school ?

**Grammar: Used to, Be used to, Get used to****1-Used to**

**Used to + verb infinitive** refers to a habit or state in the past. It is used only in the past simple tense.

We also say **used to** to express a state/an action that existed/occurred in the past but doesn't exist now.

Eg. *I **used to walk** to school every day when I was a child.*

*I **used to believe** in magic when I was a child.*

Structure of questions:

**did(n't) + subject + use to be**

***Did(n't) he use to work** in your office?*

Structure of negative:

**subject + didn't + use to be.**

***I didn't use to like** wine, but now I love it.*

**2-Be used to/Get used to**

- A. **Be used to** is used to express that a situation is not new or strange, or is no longer new or strange.

Eg.

-I've lived here for ten years now so I'm **used to** driving in the city.

-He's **not used to** working at night so he sometimes falls asleep.

-Are you **used to** the climate?

-I **wasn't used to** working such long hours when I started my new job.

### **be used to + noun/gerund**

Negative: **be not used to.**      *I don't understand him: I'm **not used to** his accent yet.*

B. **Get used to** is used to express that an action/situation becomes less strange or new, or becomes more comfortable.

Eg.

-It took them a long time to **get used to** their new boss.

-Have you **got used to** driving on the left yet?

-She **is getting used to** waking up early for her new job.

### **get used to + noun/gerund**

## Mathematical tools: ASSOCIATION SIGNS, SINES, AND COSINES

### **Association Signs**

1( ) Parenthesis (ses)

2[ ] Brackets

3{ } Braces

Example problems

1  $a(a + b)$     a into b plus c; alternatively we say:  
a into open parenthesis c plus b close parenthesis

2  $d\{c + [e + \langle h + \dots \rangle]\}$     We say: d into open braces c plus open  
brackets e plus open parenthesis h plus dot dot close parenthesis  
close brackets close braces

### **Sines and Cosines**

| English   | French     |
|-----------|------------|
| Sine      | Sinus      |
| Cosine    | Cosinus    |
| Tangent   | Tangente   |
| Cosecant  | Cosécant   |
| Secant    | Sécant     |
| Cotangent | Cotangente |

**TRANSLATION:** Translate the passage in italic into French

## Session9-Study of scientific text: Applied Physics

### VOCABULARY:

**Explain the underlined words through definition, synonym, antonym or translation.**

eg.

continuum: A continuous series or whole

to overlap: to have something in common (with)

interdisciplinary: involving two or more academic, scientific, or artistic disciplines

symptomatic of: If something (bad) is symptomatic of something else, it is **caused by** the other thing and is proof that it exists

... etc

### Text:

Applied physics is a term for physics research that combines «Pure» physics with engineering. **Pure physics** is the study of the basic physical properties of matter, and all that derives from it, such as energy and motion. **Applied physics** uses this same line of investigation to solve technological problems. It may be easy to identify research as «applied» or «pure» in cases where a direct practical application is being sought after. For instance Einstein's special theory of relativity is pure physics, and designing fiber optic technology is applied. The distinction between the two may be more **blurred**, however. Certainly, there is a **continuum** of research topics along the spectrum between applied and pure. But to be considered applied, the research must at least be concerned with the potential technological or practical applications of their research, if not directly engaged in solving an engineering problem. Applied physics research may be concerned with developing instrumentation for scientific research. Indeed, much of the **instrumentation** used by physics researchers is so advanced that it is custom built by the researchers themselves. High-energy physicists working on particle accelerators like the European Organization for Nuclear Research (CERN) are a good example of physicists who build their own instrumentation. Applied physics, as an academic discipline, is a relatively new invention with a somewhat small number of universities having departments in the field. Often, a department of applied physics will draw faculty from the physics department and engineering departments of a university. It is **common** for the faculty to hold joint appointments in more than one department. ***There is a growing trend towards interdisciplinary research in all scientific fields, and the formalized overlap of engineering and physics research in the form of applied physics departments at***

*universities is symptomatic of this trend. There are a wide variety of research topics that may be considered to be applied physics.* One example is the development of **superconductors**. A superconductor is a material that will conduct electricity without resistance below a certain temperature. Superconducting **magnets** are essential for the function of magnetic resonance imaging (MRI) machines, particle accelerators, and nuclear magnetic resonance (NMR) **spectrometers**. Research into the physical properties and theory behind superconducting magnets would properly be considered pure physics. Attempts to build improved superconductors, and to find new applications for them would certainly be considered to be applied physics. Other well known examples of this type of research include photovoltaics and **nanotechnology**.

## Grammar: UNLESS CLAUSES

### Presentation:

1-If I get money, I will build a house → Unless I get money, I won't build a house.

2-If you don't leave now, you will miss the bus → Unless you leave now, you will miss the bus.

3-If she doesn't see the goods, she won't pay → Unless she sees the goods, she won't pay.

### Formation:

If + Affirmative + Affirmative =====→ Unless + Affirmative + Negative

If + Negative + Affirmative =====→ Unless + Affirmative + Affirmative

If + Negative + Negative =====→ Unless + Affirmative + Negative

## Mathematical Tools: DIFFERENTIAL AND INTEGRAL CALCULUS

### (a) Derivatives

1  $\frac{dy}{dx}$  = The first derivative of y with respect to x

### (b) Integrals

1  $Z = \int f(x)dx$  Z equals integral f x dx (indefinite integral)

2  $A_b = \int_{-k}^{+k} (c - y)dx$  A sub b equals integral from minus k to plus k c minus y dx

## Session10-Reading Comprehension

Read the text carefully and answer the questions.

### QUESTIONS

- 1)-What is applied physics?
- 2) Applied physics is also Pure physics? True/ False
- 3)- Give one example of research topics that may be considered to be applied physics.

### Grammar: The use of Both, either, neither

We use **either**, **neither** and **both** when we are talking about two things.

**either** = one OR the other → *Would you like tea or coffee? ~Either; I don't mind.*  
(= 'You can give me tea OR coffee; I have no preference.')

**neither** = not one and not the other → *Would you like ham or beef in your sandwich? ~Neither; I'm a vegetarian.*  
(= 'I don't want ham and I don't want beef.')

**both** = the first AND the second → *I take both milk and sugar in my coffee.*  
(= 'I take sugar. I also take milk.')

We use **either** with a **singular noun**.  
We use **either of** with **aplural noun**.  
We use **asingular verb** with **either** and **either of**.  
*either car*  
*either of the cars*  
*Either day is fine for me.*  
*Either of the days is fine for me.*

We use **neither** with **asingular noun**.  
We use **neither of** with **aplural noun**.  
We use **asingular verb** with **neither** and **neither of**.  
*neither house*  
*neither of the houses*  
*Neither day was suitable.*  
*Neither of the days was suitable.*

We use **both** with **aplural noun**.  
*both houses*  
*both of the houses*  
*Both (of) my brothers are tall.*

We use **of** before the pronouns *us*, *you*, *them*.  
*both of us*, *both of you*, *either of them*, *neither of them*, etc.

Between *of* and a noun we use *these*, *those* or *my*, *your*, *John's*, etc., or *the*.  
*both of those houses*, *neither of my brothers*,  
*both of John's sisters*, *either of the cinemas*

Negative + Negative + **either** ↔ **neither ....nor**

- **either . . . or**  
We can go to **either** Greece **or** Spain for our holiday.  
It's my final offer – you can **either** take it **or** leave it.
- **both . . . and**  
**Both** rugby **and** football are popular in France.
- **neither . . . nor**  
eg. Marriage is **neither** heaven **nor** hell, it is simply purgatory. (Abraham Lincoln)

## Reading Molecular Formulas and Equations (Part1)

### MOLECULAR FORMULAS

- 1 **O** Oxygen atom
- 2 **O<sub>2</sub>** O two (Oxygen molecule)
- 3 **O<sub>3</sub>** O Three Ozone
- 4 **H<sub>2</sub>** H two Hydrogen
- 5 **H<sub>2</sub>O** H two O Water
- 6 **H<sub>2</sub>SO<sub>4</sub>** H two S O four Sulphuric Acid
- 7 **FeO** FeO (Ferrous Oxide)
- 8 **Fe<sub>2</sub>O<sub>3</sub>** Fe two O three (Ferric Oxide)

**TRANSLATION:** Translate the passage in italic into French

## Session11-Study of scientific text: Physics Jobs

### VOCABULARY:

**Explain the underlined words through definition, synonym, antonym or translation.**

eg.

pave the way : make possible

clinical trials ==► des essais cliniques

simulation software: ==► logiciel de simulation

tidal wave: ==► raz de marée

internship: ==► stage

...etc

**Text:**

A career in science and research can be an exciting opportunity for individuals with a background in physics. These professionals work with innovative technology, and their research **paves the way** to innovation ranging from planetary discoveries to disaster prevention. There are a variety of physics jobs available ranging from positions in laboratories to commercially funded projects. The most common job for someone with a physicist degree is that of a physicist. Some physicists work at government facilities, researching nuclear physics and reactors. Military physicists are responsible for tasks such as analyzing unexplained phenomena and developing weapons for space. **Research scientists** are another choice for physics jobs and are responsible for performing scientific experiments and analyzing research studies. These scientists perform investigations in a laboratory that may lead to discoveries in areas such as medicine and the environment. Other duties include **designing clinical trials** and writing scientific documents. Most facilities also require the scientist to submit documentation necessary to get funding for additional research projects. Astrophysicists, also known as astronomers, have a different category of physics jobs. An astrophysicist studies **constellations** and other planetary matter. Some tasks this scientist may be responsible for are studying the properties of nearby galaxies by using data from satellites and reviewing images from computerized telescopes. The majority of astrophysicists in the United States are employed with the National Aeronautics and Space Administration (NASA), the military or planetariums. Astrophysicists also use mathematical equations and physics to develop theories about space and the solar system. ***One of the physics jobs that use a combination of physics and seismology is that of a geophysicist. A geophysicist uses physics principles to study the earth's movement and the planetary system. This researcher analyzes earthquakes and seismic movement to predict future disasters.*** Other responsibilities of this person include studying the planets' electromagnetic pull and using **simulation software** to simulate volcanic activity and **tidal wave** forces. Physicist professors prepare students for physics jobs by teaching classes in astronomy, orbital mechanics and physics. Professors teach undergraduate classes as well as courses for advanced degree programs. Instructors also demonstrate experiments to students and administer exams to students. Some professors represent the department at faculty meetings and write reports for physics lectures. Individuals who are interested in any number of the available physics jobs should get a degree in physics and take advanced classes in their specific field of interest. Most agencies require several years of laboratory experience. Students are encouraged to seek an **internship** with a governmental agency or research facility in order to meet this requirement.

## Grammar: INVERSION

## When do we use inversion?

We use inversion in questions. But we also sometimes use inversion in other cases, when we are not asking a question.

1: When we use a negative adverb or adverb phrase at the beginning of the sentence.

- **Seldom** have I seen such beautiful work.  
(‘**Seldom**’ is at the beginning, so we use inversion. This sentence emphasizes what beautiful work it is.)
- I have seldom seen such beautiful work.  
(‘**Seldom**’ is in the normal place, so we don’t use inversion. This is a normal sentence with no special emphasis.)

We only use inversion when the adverb modifies the whole phrase and not when it modifies the noun: Hardly anyone passed the exam. (No inversion.)

**A-Here are some negative adverbs and adverb phrases that we often use with inversion: never, seldom, rarely, only then, not only ... but, no sooner, scarcely, only later, nowhere, little, only in this way, in no way, on no account.**

Hardly **had I got** into bed when the telephone rang.

Only then **did I understand** why the tragedy had happened.

No sooner **had we arrived** home than the police rang the doorbell.

On no account **should you do** anything without asking me first.

Little did he know!

**B-In the following expressions, the inversion comes in the second part of the sentence: not until, not since, only after, only when, only by.**

Only after I’d seen her flat **did I understand** why she wanted to live there.

Only when we’d all arrived home **did I feel** calm.

Not until I saw John with my own eyes **did I really believe** he was safe.

**2- We can use inversion instead of ‘if’ in conditionals with ‘had’ ‘were’ and ‘should’.**

**This is quite formal:**

- Normal conditional: If I had been there, this problem wouldn’t have happened.
- Conditional with inversion: **Had I been** there, this problem wouldn’t have happened.

**3- We can use inversion if we put an adverbial expression of place at the beginning on the sentence.**

This is also quite formal or literary:

- On the table **was all the money** we had lost. (Normal sentence: All the money we had lost was on the table.)
- Round the corner **came the knights**. (Normal sentence: The knights came round the corner.)

**4- We can use inversion after ‘so + adjective ... that’:**

- So beautiful **was the girl** that nobody could talk of anything else. (Normal sentence: the girl was so beautiful that nobody could talk of anything else.)

## Reading Molecular Formulas and Equations (Part2)

### CHEMICAL EQUATIONS

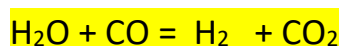
9  $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$  Hydrogen plus Oxygen gives Water. The equation is unbalanced.

10  $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$  2 H two plus O two give two H two O.

Two molecules of Hydrogen plus one molecule of Oxygen give two molecules of Water.

Equilibrium Reactions

Example of an Equilibrium Equation



H two O plus CO in equilibrium with H two plus CO two

## Session12-Reading comprehension

**Read the text carefully and answer the questions.**

### QUESTIONS

1-Name four jobs a physicist can do.

2-What are the tasks of: a)-a military physicist ? b)-an astrophysicist ?

3- How do geophysicists study the planetary system ?

4-What are the requirements to get physics jobs ?

### Grammar: PHRASAL VERBS

#### DEFINITION

A phrasal verb is a group of words that acts like a verb and consists of :

- 1 A verb followed by a preposition  
eg. to look into = investigate  
To look after = take care of (prendre soin de)
- 2 A verb followed by an adverb  
eg. to throw away = jeter
- 3 A verb followed by a preposition and an adverb  
eg. to put up with (somebody or something) = tolérer (tolerate)

Phrasal verbs have their own meanings which are different from the meanings of the verbs from which they are formed.

Phrasal verbs form an important part of the scientific literature in English.

### EXAMPLES OF PHRASAL VERBS

Consist of = être composé de, être constitué de

Consist in = consister à

Lead to = aboutir à

Result in = aboutir à

Result from = provient de

Focus on = se concentrer sur

Depend on = dépendre de

Point out = faire remarquer

Lead to = aboutir à

Base on = se baser sur

Carry out = effectuer

React with = réagir avec

Refer to = se référer à

Account for = expliquer

Dream up = imaginer concevoir

Break into = subdiviser en

Rely on = compter sur, dépendre de, avoir confiance en

Deal with = avoir affaire à, traiter avec

Prevent from = empêcher quelqu'un de faire quelque chose

Figure out = calculer, arriver à comprendre

Come up with = proposer, suggérer

Suffer from = souffrir de (eg. une maladie)

### EXAMPLE SENTENCES WITH PHRASAL VERBS

- Point out** eg. Koffi pointed out the errors in Koku's homework.
- Carry out** eg. Some experiments carried out by scientists did not result in success.
- Dream up** eg. The professor advises his students to dream up new ways of solving their problems
- Break into** eg. White light can be broken into several colours.
- Consist of** eg. An atom consists of electrons, neutrons and protons.
- Result in** eg. The driver's carelessness resulted in a road accident.
- Result from** eg. The road accident resulted from the driver's carelessness.
- Suffer from** eg. My sister often suffers from malaria.
- Prevent from** eg. The market women prevented the thief from running away.
- Depend on** eg. Success in an exam depends on hard work.
- Figure out** eg. It has taken three or more centuries to figure out some mathematical enigmas
- Lead to** eg. Experiments carried out by scientists may lead to success or failure.
- Focus on** eg. Geologists mainly focus on the study of rocks.
- Account for** eg. Theory of universal gravitation was proposed to account for the observed movement of the planets.
- React with**
- eg. Acids react with bases to give salt and water.
- Zinc reacts with hydrochloric acid to give hydrogen gas.
- Come up with** eg. The student came up with a new idea for solving the problem.
- Consist in**
- eg. The procedure consists in adding acid to the metal and neutralizing excess acid with sodium hydroxide.
- Refer to** eg. Natural science is sometimes referred to as 'hard science'.
- Deal with** eg. We won't deal with that man because he is dishonest.
- Rely on** eg. John relied on the train being on time, but he was disappointed.

**EXERCISES ON PHRASAL VERBS**

COMPLETE THE FOLLOWING SENTENCES WITH THE APPROPRIATE PREPOSITION.

- 1 Oxygen reacts ..... hydrogen to give water
- 2 This discovery led ..... the development of new technologies.
- 3 It prevents products ..... reacting with the steel.
- 4 The amount of solar energy received by a region depends..... the distance from the equator.
- 5 The laboratory scientist carried ..... an experiment on white mice.
- 6 A beam consists ..... horizontal member resting on a vertical support at each end.
- 7 She came up ..... a new idea.
- 8 Science requires measurements to work ..... problems and find rules.
- 9 The theoretician ..... his arguments ..... three points.
- 10 Anaesthetics are divided ..... two broad groups: general anaesthetics and local anaesthetics.
- 11 The researcher could not account ..... the mistake he had made in the calculation.
- 12 If you take three ..... ten you are left with seven.

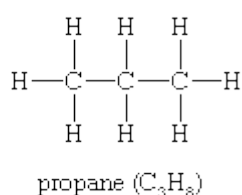
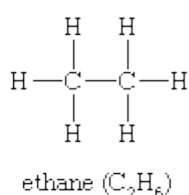
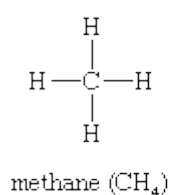
**Reading Molecular Formulas and Equations (Part3)****SOME EXPRESSIONS IN CHEMISTRY**

- 1 the reaction takes place =la reaction se produit
- 2 the reaction produces H two O = la reaction produit H<sub>2</sub>O
- 3 Undergo a reaction =subir une reaction
- 4 Initiate a reaction = déclencher une reaction
- 5 To balance an equation = equilibrer une equation
- 6 Mendeleev Table = Tableau de Mendeleev
- 7 Analysis = Analyze
- 8 To analyze = analyser
- 9 Synthesis = Synthese
- 10 To synthesize = Synthétiser

**The Hydrocarbons**

- 1-  $\text{CH}_4$  CH four (methane)
- 2-  $\text{C}_2\text{H}_6$  C two H six or  $\text{CH}_3 - \text{CH}_3$  CH three single bond CH three (ethane)
- 3-  $\text{C}_3\text{H}_8$  C three H eight (propane)
- 3-  $\text{C}_2\text{H}_4$  C two H four.  $\text{CH}_2$  double bond  $\text{CH}_2$  (ethene or ethylene)
- 4-  $\text{C}_2\text{H}_2$  C two H two. CH triple bond CH (acetylene)

### MOLECULAR STRUCTURE



**TRANSLATION:** Translate the passage in italic into French